

## Лабораторная работа №11

### 1. Постройте кубический сплайн;

#### Варианты заданий:

1.  $y = e^x$ ,  $x_i = 0.1 * i$ ;  $i = 0, \dots, 10$ ; значения функции в этих точках  $y_i = e^{x_i}$ . Для интерполирования имеем узлы  $x_j = 0.05 * j$ ,  $j = 0, \dots, 20$ .
2.  $y = \sin(x)$ ,  $x_i = 0.2\pi * i$ ,  $i = 0, \dots, 10$ ;  $x_j = 0.1\pi * j$ ,  $j = 0, \dots, 20$ .
3.  $y = \cos(x)$ ,  $x_i = 0.2\pi * i$ ,  $i = 0, \dots, 10$ ;  $x_j = 0.1\pi * j$ ,  $j = 0, \dots, 20$ .
4.  $y = x^3 + 1$ ,  $x_i = 0.2 * i$ ,  $i = 0, \dots, 10$ ;  $x_j = 0.1 * j$ ,  $j = 0, \dots, 20$ .
5.  $y = 5 + x^2$ ,  $x_i = 0.2 * i$ ,  $i = 0, \dots, 10$ ;  $x_j = 0.1 * j$ ,  $j = 0, \dots, 20$ .
6.  $y = 10 + 2x + e^x$ ,  $x_i = 0.1 * i$ ,  $i = 0, \dots, 10$ ;  $x_j = 0.05 * j$ ,  $j = 0, \dots, 20$ .
7.  $y = x^4 - 5x^3 + 3x^2 + x - 2$ ,  $x_i = 0.2 * i$ ,  $i = 0, \dots, 10$ ;  $x_j = 0.1 * j$ ,  $j = 0, \dots, 20$ .
8.  $y = x + 1 + \ln(x)$ ,  $x_i = 0.2 * i$ ,  $i = 0, \dots, 10$ ;  $x_j = 0.1 * j$ ,  $j = 0, \dots, 20$ .
9.  $y = x^3 - 2x^2 + 5x + 2$ ,  $x_i = -1 + 0.2 * i$ ,  $i = 0, \dots, 20$ ;  $x_j = -1 + 0.1 * j$ ,  $j = 0, \dots, 40$ .
10.  $y = e^x + 1/(1+x)$ ,  $x_i = 0.2 + 0.5 * i$ ,  $i = 0, \dots, 10$ ;  $x_j = 0.2 + 0.25 * j$ ,  $j = 0, \dots, 20$ .
11.  $y = 1 + \sin(x) + \cos(x)$ ,  $x_i = -3\pi/4 + 0.1\pi * i$ ,  $i = 1, \dots, 20$ ;  
 $x_j = -3\pi/4 + 0.15\pi * j$ ,  $j = 1, \dots, 10$ .
12.  $y = 5 + 1/(\sin(x) + \cos(x))$ ,  $x_i = -\pi/4 + 0.1\pi * i$ ,  $i = 1, \dots, 9$ ;  
 $x_j = -\pi/4 + 0.05\pi * j$ ,  $j = 1, \dots, 18$ .
13.  $y = 5 + 2^x$ ,  $x_i = 0.1 * i$ ,  $i = 0, \dots, 20$ ;  $x_j = 0.05 * j$ ,  $j = 0, \dots, 40$ .
14.  $y = 6 + \lg(x)$ ,  $x_i = 2 + 0.1 * i$ ,  $i = 0, \dots, 10$ ;  $x_j = 2 + 0.05 * j$ ,  $j = 0, \dots, 20$ .
15.  $y = 2 + 1/(1+x)$ ,  $x_i = 0.3 + 0.5 * i$ ,  $i = 0, \dots, 10$ ;  $x_j = 0.3 + 0.25 * j$ ,  $j = 0, \dots, 20$ .
16.  $y = 5 + 2\sin(x)/3$ ,  $x_i = 0.1\pi * i$ ,  $i = 0, \dots, 10$ ;  $x_j = 0.05\pi * j$ ,  $j = 0, \dots, 20$ .
17.  $y = 7 + 3\cos(x)/4$ ,  $x_i = 0.1\pi * i$ ,  $i = 0, \dots, 10$ ;  $x_j = 0.05\pi * j$ ,  $j = 0, \dots, 20$ .
18.  $y = 2x^2 + 3x - 10$ ,  $x_i = 0.2i - 1$ ,  $i = 0, \dots, 10$ ;  $x_j = 0.1 * j - 1$ ,  $j = 0, \dots, 20$ .
19.  $y = 10 + (2x + \ln(x))/2$ ,  $x_i = 0.2 + 0.1 * i$ ,  $i = 0, \dots, 10$ ;  $x_j = 0.2 + 0.05 * j$ ,  $j = 0, \dots, 20$ .
20.  $y = x^3 + 3x^2 - 3x + 10$ ,  $x_i = -1 + 0.2 * i$ ,  $i = 0, \dots, 30$ ;  $x_j = -1 + 0.25 * j$ ,  $j = 0, \dots, 24$ .
21.  $y = 2 + 10e^{-x}$ ,  $x_i = 0.3 + 0.5 * i$ ,  $i = 0, \dots, 10$ ;  $x_j = 0.3 + 0.25 * j$ ,  $j = 0, \dots, 20$ .
22.  $y = 7 + 3^x/10$ ,  $x_i = 1 + 0.1 * i$ ,  $i = 0, \dots, 10$ ;  $x_j = 1 + 0.05 * j$ ,  $j = 0, \dots, 20$ .
23.  $y = 5 + 3x + \lg(x)$ ,  $x_i = 2 + 0.1 * i$ ,  $i = 0, \dots, 10$ ;  $x_j = 2 + 0.05 * j$ ,  $j = 0, \dots, 20$ .
24.  $y = ex + x^2 + x + 1$ ,  $x_i = -1 + 0.2 * i$ ,  $i = 0, \dots, 20$ ;  $x_j = -1 + 0.1 * j$ ,  $j = 0, \dots, 40$ .
25.  $y = 2x + x^2 + (\sin(x) + \cos(x))/0.5$ ,  $x_i = 0.2\pi * i$ ,  $i = 0, \dots, 10$ ;  $x_j = 0.1\pi * j$ ,  $j = 0, \dots, 20$ .

Задача для тестирования алгоритма прогонки:

$$\begin{cases} -5x_1 + 2x_2 = -3, \\ -3x_1 + 8x_2 - 4x_3 = 1, \\ 4x_2 - 10x_3 + 4x_4 = -2, \\ x_3 + 5x_4 - 2x_5 = 4, \\ x_4 - 6x_5 - 4x_6 = -9, \\ x_5 + 4x_6 + 2x_7 = 7, \\ x_6 - 3x_7 = -2. \end{cases} \quad x = (1, 1, 1, 1, 1, 1, 1).$$